

The actual quadratic source and integrated cubic residual

HOCF–RIESZ, Round 015

18 September 2026

Status. THM–036 and THM–037 have complete sealed construction and fresh reconstruction; AUD–049 compares the entire conjunction. Separate fresh hostile review AUD–050 also passes. Its three damaged fraction command prefixes have a separately sealed rendering clarification, with no mathematical change and all original bytes preserved. This synthesis records their exact conditional implications; it does not certify a fluctuation law or the higher hierarchy. The complete proofs and source qualifications are in the separately sealed Round 015 memoranda and audit packets.

The actual energy controls a positive omitted part

Fix $d \geq 4$, $0 < s < 2$, finite T , a fixed smooth real terminal test h , and $0 \leq \nu \leq \nu_* < \infty$. Use the coefficient-one periodic Riesz kernel on the unit Haar torus, $K = -\nabla g$, actual particles with drift $N^{-1} \sum_{j \neq i} K(X_i - X_j)$ and noise $\sqrt{2\nu}$, initially iid Haar. Let f_t be the actual backward Fourier test and

$$J_t(x, y) = K(x - y) \cdot (\nabla f_t(x) - \nabla f_t(y)), \quad P_N[F] = \frac{1}{2N^2} \sum_{i \neq j} F_{ij} - \frac{1}{N} \sum_i \int F(X_i, y) dy + \frac{1}{2} \int F.$$

All singular kernels use only off-diagonal empirical evaluations and genuine Haar contractions. Put $\alpha = (d - s)/2 > 1$ and $A = 4^\alpha \pi^{d/2} / \Gamma(s/2)$. The positive heat representation is

$$g(z) = A \int_0^\infty t^{\alpha-1} (p_t(z) - 1) dt.$$

For $r > 0$ set

$$g_r = A \int_r^\infty t^{\alpha-1} (p_t - 1) dt, \quad H_r = A \int_0^r t^{\alpha-1} p_t dt \geq 0, \quad c_r = Ar^\alpha / \alpha.$$

Then $g = g_r + H_r - c_r$ off zero. With $D_2 = N^{-2} \sum_{i \neq j}$, $H_N = N^{-1} \sum_{i < j} g_{ij}$ and $a_r(k) = \widehat{g}_r(k) > 0$ for $k \neq 0$,

$$\mathcal{E}_r = \sum_{k \neq 0} a_r(k) |\widehat{\eta}_N(k)|^2 \geq 0, \quad \mathcal{E}_r + D_2[H_r] = \frac{2H_N}{N} + \frac{g_r(0)}{N} + \left(1 - \frac{1}{N}\right) c_r.$$

The actual energy satisfies $\mathbb{E}H_N(t) \leq 0$: at positive noise use smooth free-energy dissipation, nonnegative entropy and the fixed- N same-noise heat passage with shifted Fatou; at zero noise use deterministic energy decrease. Since $g_r(0) \leq Cr^{-s/2}$,

$$\mathbb{E}\mathcal{E}_r + \mathbb{E}D_2[H_r] \leq C(N^{-1}r^{-s/2} + r^\alpha).$$

The different labelled-pair coefficient is $\mathbb{E}H_r(X_1 - X_2) \leq g_r(0)/(N - 1) + c_r$. The energy self subtraction is smooth; no singular diagonal value is assigned.

A summable Fourier commutator bound

For each smooth test v let $J_r[v]$ use $-\nabla g_r$ in place of K . The Gaussian estimate, translate by translate, is

$$\text{dist}(z, 0) |\nabla p_t(z)| \leq C p_{2t}(z).$$

Thus $|J[v] - J_r[v]| \leq C \|D^2 v\|_\infty H_{2r}$. Retaining all three parts of P_N yields an absolute tail bound by $C \|D^2 v\|_\infty (N^{-1} r^{-s/2} + r^\alpha)$.

For a nonzero terminal mode $n = k + l$, the exact pair Fourier coefficient is

$$-4\pi^2 n \cdot (ka_r(k) + la_r(l)).$$

For $k, l \neq 0$ and $r|n|^2 \leq 1$, direct differentiation of the heat integral, followed by a split at $|k| = 2|n|$, gives

$$|n \cdot (ka_r(k) + la_r(l))| \leq C(1 + |n|)^{2\alpha+2} \sqrt{a_{r/32}(k)a_{r/32}(l)}.$$

Indeed $\|D_v(va_r(v))\| \leq C|v|^{-2\alpha}e^{-2\pi^2 r|v|^2}$, while $a_r(v) \geq c|v|^{-2\alpha}e^{-4\pi^2 r|v|^2}$. The far segment stays away from zero; the near nonzero integer frequencies have norms at least one. The slack cutoff absorbs the Gaussian shift with only polynomial dependence on n .

The smooth source has $J_r[v](x, x) = 0$ exactly, so $P_N[J_r[v]] = \langle \rho_N^{\otimes 2}, J_r[v] \rangle / 2$. Weighted Cauchy–Schwarz in k bounds the corresponding quadratic expression by $C(1 + |n|)^{2\alpha+2} \mathcal{E}_{r/32}$. Choosing

$$r_{N,n} = N^{-2/d}(1 + |n|)^{-2}$$

inside this estimating argument and retaining the omitted positive part gives

$$\mathbb{E}|P_N[J[e_n]]| \leq C(1 + |n|)^{d+2} N^{s/d-1}.$$

The actual backward coefficients have modulus at most $|\widehat{h}(n)|$. Their weighted sum is finite for fixed smooth h . Fourier partial sums converge in C^2 ; the actual pair w_s moment is uniformly bounded by the energy and the local coefficient-one comparison. These facts identify the summed estimate with the genuine singular source in actual L^1 . Hence

$$\sup_{0 \leq t \leq T} \mathbb{E}|P_N[J_t]| \leq CN^{s/d-1}.$$

This is THM–037, uniform in the bounded noise interval, including zero. Writing $b = \min(1, \nu^{-1})$ for positive noise and $b = 1$ at zero,

$$\sqrt{Nb} \mathbb{E} \int_0^T |P_N[J_t]| dt \leq C\sqrt{b} N^{s/d-1/2} \longrightarrow 0.$$

The constructor uses an alternative low/high terminal-frequency split with cutoff $r/64$ and a finite Fourier seminorm of order $d + 4$; the complete fresh reconstruction above instead uses the mode-dependent cutoff. Both prove the same frozen uniform bound without an exponentially growing test cost.

The exact integrated cubic conclusion

Now take positive finite microscopic criticality

$$\lambda_N = \beta_N N^{s/d-1} \rightarrow \lambda \in (0, \infty), \quad \nu_N = \beta_N^{-1}, \quad \theta = 1 - s/d, \quad p = s + 2, \quad a = s/p.$$

Then $b_N = 1$ eventually and $\nu_N N^{2/p} = \lambda_N^{-1} N^{s(s+2-d)/(dp)} \rightarrow 0$. Let Φ be the genuine symmetric terminal-zero pair inverse with both coefficient-one response slots. Define

$$C\Phi = \text{Sym}_3[K(x-z) \cdot \nabla_x \Phi(x, y)], \quad B\Phi = K(x-y) \cdot (\nabla_x - \nabla_y)\Phi,$$

where Sym_3 averages all six permutations. The accepted R8 domain supplies the genuine singular identity and true square-integrable martingale. Literal ordered-label counting gives

$$dP_N[\Phi_t] = \left[-P_N[J_t] + U_3[C\Phi_t] + \frac{1}{N} \rho_t[(B\Phi_t)_\mu] + \frac{1}{2N} \int B\Phi_t \right] dt + dM_t.$$

The cubic U_3 has denominator N^3 and every Haar contraction; its background terms remain even at $N = 2$. Terminal zero gives

$$\begin{aligned} \int_0^T U_3[C\Phi_t] dt &= \int_0^T P_N[J_t] dt - P_N[\Phi_0] - M_T \\ &\quad - \frac{1}{N} \int_0^T \rho_t[(B\Phi_t)_\mu] dt - \frac{1}{2N} \int_0^T \int B\Phi_t dt. \end{aligned}$$

Each term is estimated in its genuine law and domain. The supplied full inverse has uniformly bounded Haar L^2 norm because $2s < d$. Exact initial iid counting gives $\sqrt{Nb_N} \mathbb{E}[P_N[\Phi_0]] \leq CN^{-1/2}$. Set $\epsilon = \min(s, d - s - 2)/2$, $q = 1 + \epsilon$ and $e = (s - \epsilon)/p$. The admitted R12 gradient bound is $|\nabla_{x,y}\Phi| \leq CN^e w_q$. Since $s + 1 + q < d$, each lower contraction has absolute scaled integral at most $CN^{e-1/2}$, retaining its exact coefficient. The complete actual R12 bracket satisfies

$$\sigma_N^2 \mathbb{E}\langle M \rangle_T \leq Cb_N N^a (N^{-\theta} + \nu_N), \quad \sigma_N = \sqrt{Nb_N}.$$

Isometry gives $\sigma_N \mathbb{E}|M_T| \leq CN^{(a-\theta)/2}$ on the critical tail. Consequently THM-036 is the precise bound

$$\boxed{\sigma_N \mathbb{E} \left| \int_0^T U_3[C\Phi_t] dt \right| \leq C \left(N^{1/2-\theta} + N^{-1/2} + N^{e-1/2} + N^{(a-\theta)/2} \right) \rightarrow 0.}$$

All exponents are strictly negative for $d \geq 4$, $0 < s < 2$; the last uses $s(s+2) < 2d$. Constants may depend on the fixed data and bounds for the critical coupling on its eventual tail. The absolute value remains after time integration. There is no claim about the integral of the absolute cubic, an evolved iid endpoint, a broader singular range, or a full limiting law. The old energy-floor and full microscopic subcritical conditions remain separate from the critical condition used here.